Mortality Prediction in ICU Patients Using Structured MIMIC-IV Features

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Abstract

This report documents and interprets the results contained in the project training notebook for ICU mortality prediction using structured features derived from the MIMIC-IV dataset. Guided by the broader project proposal, the report addresses an initial but clinically meaningful task: binary prediction of ICU mortality from tabular demographic, administrative, vital-sign, and laboratory variables. The modeling workflow used an 80/20 train-test split, structured preprocessing with imputation and one-hot encoding, and comparison of three required model families: logistic regression, random forest, and a multilayer perceptron. Cross-validation results showed that the multilayer perceptron achieved the strongest PR-AUC among the required models (0.386), while random forest achieved the highest ROC-AUC (0.851). The notebook nevertheless selected logistic regression as the final reportable model because of its transparency and easier clinical interpretation. After hyperparameter tuning and threshold selection to target approximately 80% recall, the final tuned logistic regression achieved a held-out ROC-AUC of 0.829, PR-AUC of 0.356, recall of 0.801, precision of 0.165, and accuracy of 0.694. An exploratory HistGradientBoosting analysis, included as a supplementary extension, improved test ROC-AUC to 0.864 and PR-AUC to 0.410, supporting the proposal's expectation that modern ensemble methods may outperform transparent baselines on structured EHR data. Overall, the report demonstrates that a meaningful mortality signal is present in the current feature set, but it also highlights the importance of threshold tuning, class imbalance, data-quality checks, calibration, and future interpretability work.

1. Introduction

Early identification of patients at elevated risk of death is a central ICU decision-support problem. In the submitted project proposal, the team framed mortality prediction as a way to support timely monitoring, prioritization, and intervention using routinely collected EHR data. The proposal also emphasized that the final project should compare interpretable baselines with stronger nonlinear models, while remaining reproducible and clinically explainable..

The current training notebook implements an important first stage of that agenda. Rather than covering the full proposal scope like dynamic prediction at admission, 24 hours, and 48 hours; post-discharge mortality; subgroup fairness; survival modeling; and formal explainability, the report focuses on a single static binary task: predicting ICU mortality from a curated structured dataset and comparing multiple machine-learning pipelines. That narrower scope is still highly informative because it shows whether the current feature set already contains meaningful predictive signals and whether an interpretable baseline can perform competitively.

2. Project Scope and Alignment with the Proposal

The proposal articulated five broader research directions: dynamic in-hospital mortality prediction at admission and during the first 24 to 48 hours of ICU care; post-discharge mortality prediction; comparison of model families; interpretability and fairness analysis; and a reproducible analytic pipeline grounded in MIMIC-IV. The notebook directly addresses two of those goals. First, it tests whether structured ICU-related variables can predict mortality with useful discrimination. Second, it compares model families with different trade-offs between interpretability and nonlinear capacity.

From that perspective, the notebook should be viewed as a baseline implementation rather than a complete realization of the proposal. It operationalizes the core classification task, establishes an evaluation routine, and provides a defensible starting point for later extensions such as calibration analysis, subgroup performance review, SHAP-based interpretation, temporal feature updates, and survival-style modeling. This framing is important when interpreting the results: the notebook meaningfully advances the project, but it does not yet exhaust the proposal's full scientific scope.

3. Data Description and Preprocessing

According to the notebook's output, the modeling table contains 74,583 rows and 26 columns, including the binary outcome variable ICU_mortality. Excluding the target, the model operates on 25 predictors spanning demographics and administrative information (for example age, sex, race, insurance, language, and marital status), ICU unit information, and summary laboratory or vital-sign measurements such as heart rate, respiratory rate, oxygen saturation, mean arterial pressure, lactate, creatinine, white blood cell count, platelets, sodium, potassium, and bilirubin.

The notebook uses a conventional 80/20 train-test split, producing 59,666 training examples and 14,917 held-out test examples. Preprocessing is implemented with a ColumnTransformer. Numerical variables are median-imputed, supplemented with missingness indicators, and standardized. Categorical variables are imputed using the most frequent category and one-hot encoded. The sole boolean feature is_male is passed through directly. A notebook check confirms that all columns are handled by the preprocessing pipeline, which is a strong reproducibility practice for tabular modeling.

Table 1. Dataset and feature summary derived from the notebook outputs

Item	Value
Total observations	74,583
Total columns	26
Predictor columns	25
Train / test split	59,666 / 14,917
Held-out positive outcomes	1,073 deaths (7.2% prevalence in the test set)
Feature groups	17 numeric, 7 categorical, 1 boolean
Highest missingness	max_val_bilirubin: 54.9%; last_val_lactate: 41.0%; max_val_lactate: 41.0%

Missingness is concentrated in a small subset of laboratory measurements. Bilirubin is missing in roughly 54.9% of rows, while both lactate variables are missing in approximately 41.0% of rows. In contrast, most other numeric predictors have very low missingness, usually below 2%. The decision to include missingness indicators for numeric features is therefore well motivated because the presence or absence of a lab test can itself encode clinical information. Figure 1 summarizes this pattern.

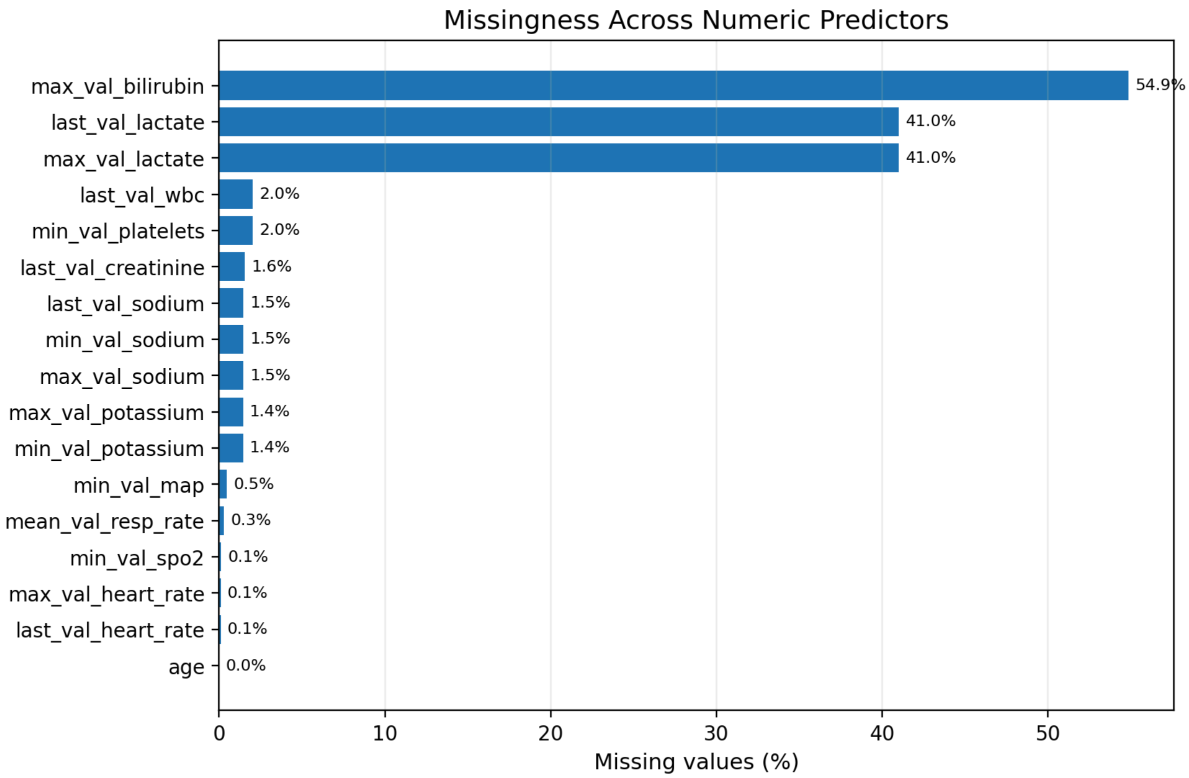

Figure 1. Missingness across numeric predictors in the modeling table. Bilirubin and lactate are the only variables with substantial incompleteness.

The descriptive statistics also expose several clinically implausible extremes in the raw feature summaries, including a maximum heart rate of 6,262, a maximum sodium value of 1,036, a minimum oxygen saturation of -8, and a minimum mean arterial pressure of -22,767. These values likely reflect measurement artifacts, unit inconsistencies, or extraction errors rather than physiologic reality.

4. Modeling Strategy and Evaluation Design

Three required supervised model families are evaluated in the notebook: logistic regression, random forest, and a multilayer perceptron classifier. These choices reflect a sensible progression from a transparent linear baseline to tree-based and neural-network alternatives. The notebook uses 5-fold stratified cross-validation on the training split and reports ROC-AUC, PR-AUC, recall, and precision. In this setting, PR-AUC is especially important because mortality is a minority outcome and ROC-AUC alone can overstate apparent performance under severe class imbalance.

One subtle but important implementation detail is that the cross-validated recall and precision values for the three candidate models are computed using each estimator's standard predict behavior, which corresponds to its default classification threshold. The final logistic-regression evaluation, however, lowers the decision threshold after tuning in order to deliberately target high recall. The notebook performs randomized hyperparameter search for logistic regression and, in an exploratory section, for HistGradientBoosting. For the chosen logistic model, cross-validated training-set probabilities are used to select the highest threshold that still achieves at least 0.80 recall. This is a clinically coherent choice for a screening-oriented mortality model, because missing a truly high-risk patient is generally more costly than reviewing a false-positive alert.

Model	ROC-AUC	PR-AUC	Recall	Precision
Logistic Regression	0.832	0.356	0.133	0.632
Random Forest	0.851	0.368	0.070	0.709
MLP Classifier	0.846	0.386	0.159	0.647

Table 2. Mean 5-fold cross-validation performance for the three required models

Table 2 shows that different models look strongest depending on the metric used. Random forest achieves the highest ROC-AUC (0.851), suggesting the best overall ranking of positives above negatives. The multilayer perceptron achieves the best PR-AUC (0.386) and the strongest default-threshold recall (0.159), while logistic regression is slightly weaker on both measures. Figure 2 visualizes the discrimination metrics directly. These results are informative for the proposal's question about model-family trade-offs: transparent models remain competitive, but nonlinear alternatives extract additional signal from the same structured inputs.

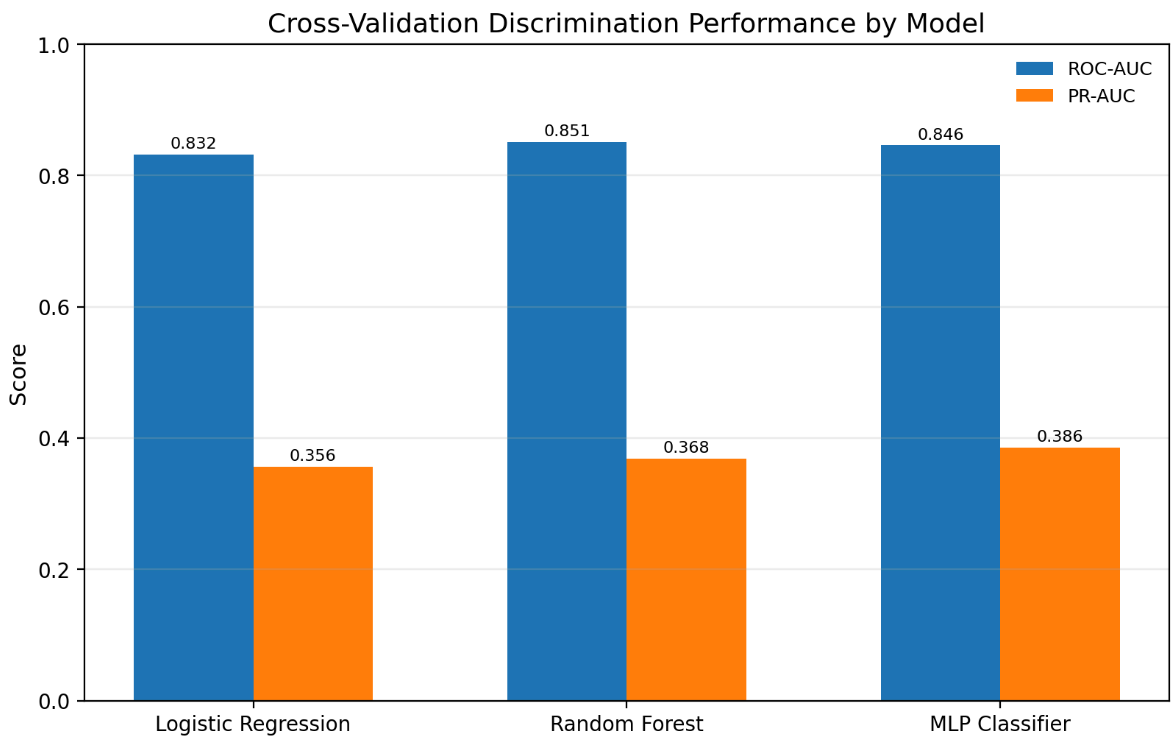

Figure 2. Cross-validation discrimination performance across the three required models. Random forest leads on ROC-AUC, while the multilayer perceptron leads on PR-AUC.

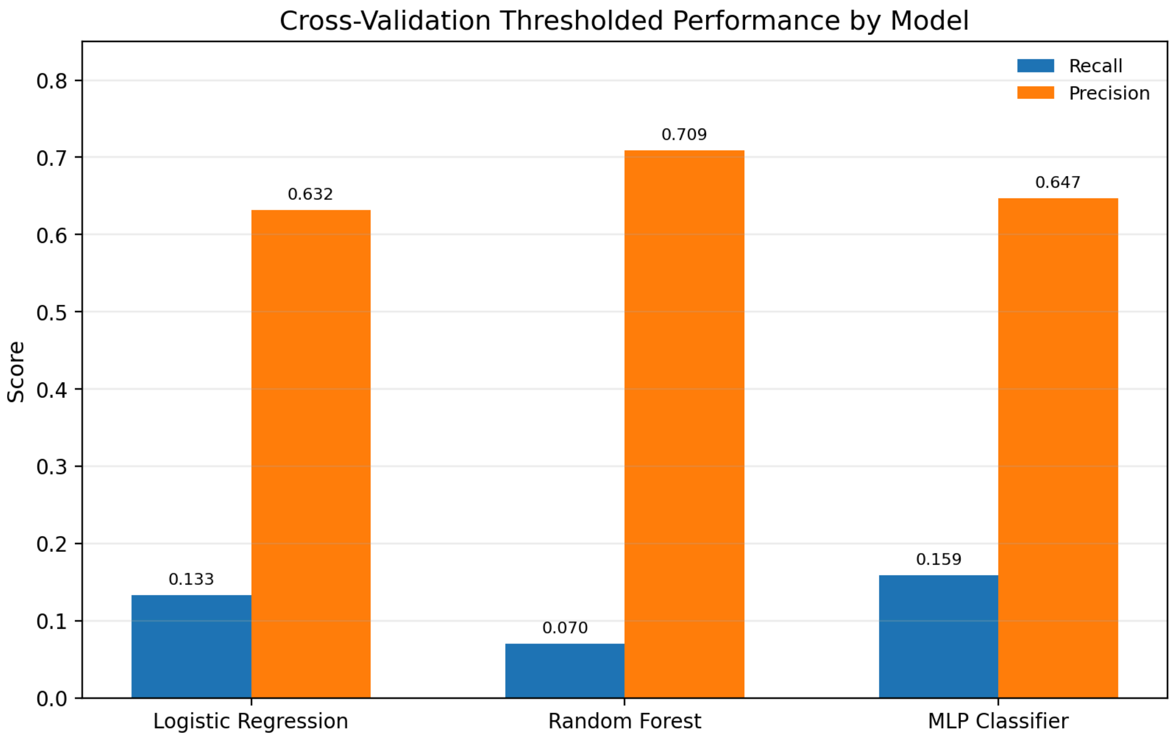

Figure 3. Cross-validation thresholded performance under each model's default prediction threshold. Random forest achieves high precision but much lower recall.

5. Results

5.1 Candidate-model comparison

For a rare-event clinical task, the contrast between ROC-AUC and PR-AUC is not just technical; it changes the substantive interpretation of performance. Random forest would appear best if the analysis stopped at ROC-AUC, yet its default-threshold recall is only 0.070, meaning it misses most deaths when converted into hard classifications. The multilayer perceptron provides the most attractive balance among the required models on PR-AUC and recall, but it is also the least transparent of the three. The notebook therefore makes a defensible project-management decision: retain logistic regression as the final reportable model because its performance is still respectable while its coefficients and linear structure are far easier to communicate to clinicians and nontechnical people.

5.2 Final tuned logistic-regression model

The tuned logistic-regression pipeline uses L2 regularization with solver = lbfgs, C = 29.764, and no class weighting, and its best cross-validated PR-AUC after randomized search is 0.356. Hyperparameter tuning does not materially change ranking performance relative to the untuned logistic baseline, which is itself an informative result: for this model family and feature set, operating-point selection matters more than small changes in regularization strength.

The thresholding step is therefore the key operational adjustment. The notebook selects a threshold of 0.0587, far below the conventional 0.50 threshold, because the explicit goal is to reach about 80% recall on cross-validated training predictions. This choice transforms the model from a conservative classifier into a screening tool designed to surface most high-risk patients at the cost of increased false-positive alert.

Metric	Value
Chosen decision threshold	0.0587
ROC-AUC	0.829
PR-AUC	0.356
Recall (sensitivity)	0.801
Precision (PPV)	0.165
Specificity	0.686
Negative predictive value	0.978
Accuracy	0.694
Confusion matrix	TN=9497, FP=4347, FN=213, TP=860

Table 3. Held-out test performance of the final tuned logistic-regression model

On the held-out test set, the final tuned logistic model achieves ROC-AUC = 0.829 and PR-AUC = 0.356. At the selected low threshold, it attains recall = 0.801, precision = 0.165, specificity = 0.686, negative predictive value = 0.978, and accuracy = 0.694. Because the test prevalence is only 7.2%, a PR-AUC of 0.356 is still meaningful, it is almost five times the baseline prevalence level that would characterize an uninformative precision-recall curve. The model therefore ranks cases substantially better than chance, even though its precision at the chosen operating point remains modest.

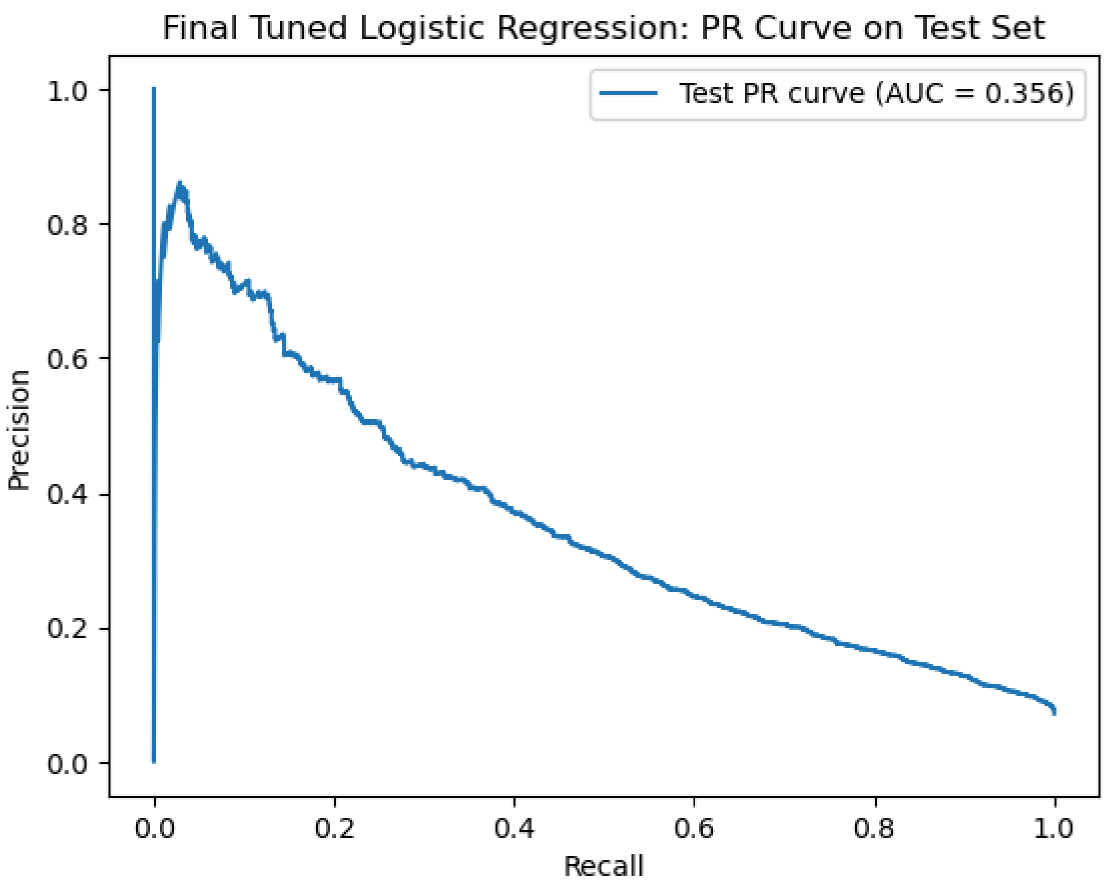

Figure 4. Precision-recall curve for the final tuned logistic-regression model on the held-out test set.

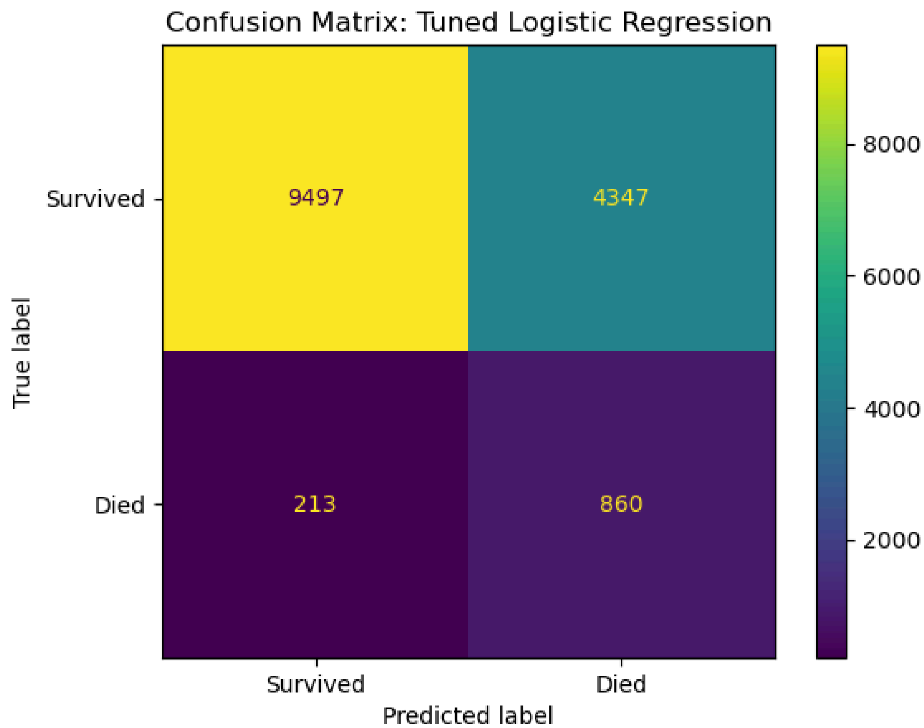

Figure 5. Confusion matrix for the final tuned logistic-regression model on the held-out test set.

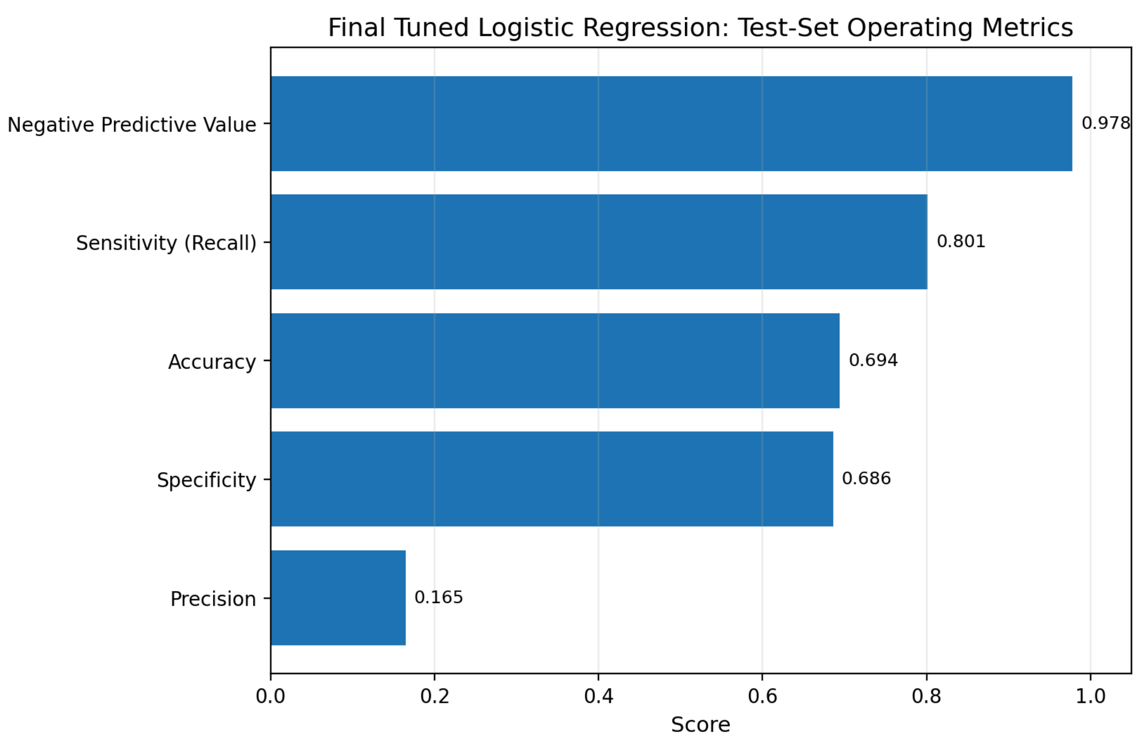

Figure 6. Test-set operating characteristics of the final tuned logistic-regression model. Recall is high by design, while precision remains limited by class imbalance and threshold choice.

The confusion matrix makes the trade-off concrete. The model identifies 860 of 1,073 deaths while missing 213, but it also generates 4,347 false positives. Put differently, the model produces about 5.1 false alarms for every correctly identified death. This burden may still be acceptable in a triage or watch-list setting, especially because the model's negative predictive value is very high, but it would likely be too imprecise for fully automated escalation decisions. The results therefore support using the model as a screening aid rather than as a standalone decision-maker.

5.3 Exploratory HistGradientBoosting extension

Although the notebook labels HistGradientBoosting as additional exploration, its results are highly relevant to the proposal's hypothesis that modern ensemble methods should outperform transparent baselines on structured EHR data. After randomized search, the exploratory model achieves best cross-validated PR-AUC = 0.398, chooses a threshold of 0.0658 to target approximately 80% recall, and delivers held-out ROC-AUC = 0.864 and PR-AUC = 0.410. At the chosen operating point, recall rises slightly to 0.816 and precision improves to 0.201.

Relative to the final logistic model, HistGradientBoosting reduces false positives by 856 while simultaneously increasing true positives by 16 at a similarly high-recall operating point. That is a practically important improvement. Figure 7 summarizes this comparison. If the project continues beyond the current notebook milestone, this exploratory result strongly suggests that gradient boosting should become the primary candidate for the proposal's advanced-model phase.

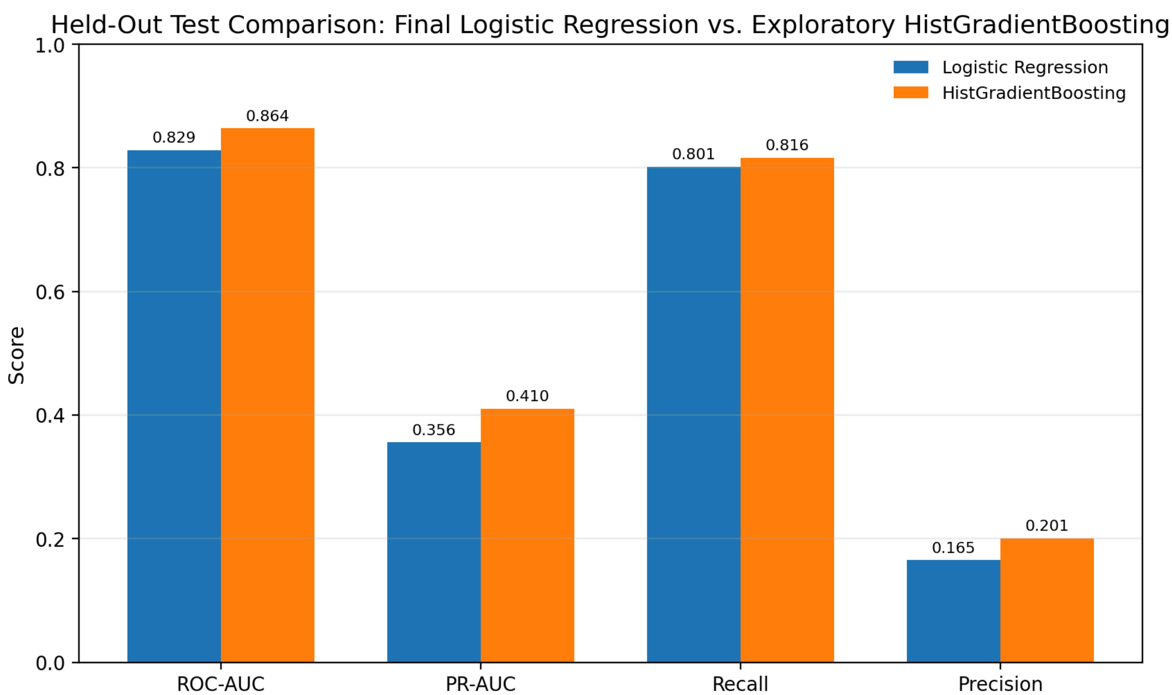

Figure 7. Held-out test comparison between the final tuned logistic-regression model and the exploratory HistGradientBoosting model. The ensemble improves every reported metric at a similar high-recall operating point.

6. Discussion

Several broader lessons emerge from the notebook. First, model ranking quality and deployment threshold are separate levers. Hyperparameter tuning barely changed logistic-regression PR-AUC, but threshold tuning dramatically altered its recall and false-positive profile. This is especially important in clinical settings, where the default 0.50 threshold is seldom optimal. Second, ROC-AUC and PR-AUC should be interpreted together. Random forest looked best on ROC-AUC among the required models, yet its low default-threshold recall would make it less attractive if the operational goal is early risk capture.

Third, the notebook supports the proposal's high-level scientific intuition. There is enough signal in the available structured features to meaningfully rank mortality risk, but more flexible nonlinear models appear able to exploit that signal better than a linear baseline. At the same time, the logistic model remains valuable because it establishes a transparent reference point and provides a simpler starting point for sensitivity analyses, and later calibration work.

Finally, the descriptive statistics underline that data quality may be a major determinant of future gains. Implausible physiologic values and heavily missing laboratory measurements can distort both linear and nonlinear models. The next performance gains may therefore come not only from more powerful algorithms, but also from better data cleaning, temporal aggregation choices, and explicit feature engineering around missingness and outliers.

7. Conclusion

The project notebook demonstrates that ICU mortality can be predicted with meaningful discrimination from the current structured MIMIC-IV feature set. As an interpretable baseline, tuned logistic regression provides a clinically sensible high-recall screening model, achieving 0.829 ROC-AUC and 0.356 PR-AUC on the held-out test set while capturing roughly 80% of deaths at a low threshold. However, the notebook's own exploratory results also show that a modern gradient-boosting ensemble can improve both discrimination and practical operating characteristics. The strongest overall conclusion is therefore twofold: the current pipeline already supports evidence-based risk stratification, and the proposal's next phase should now prioritize data-quality improvements, calibration, interpretability tooling, and advanced ensemble models to turn that baseline into a more robust clinical research asset.